

Homework 2 Solutions

BSTA 551

Problem 1: Fisher Information and CRLB

A clinical trial is testing a new treatment for hypertension. The time (in days) until a patient achieves target blood pressure is modeled as an Exponential distribution with rate parameter λ (where $E(X) = 1/\lambda$).

In a pilot study, the following times (in days) were observed for $n = 8$ patients:

$$X = \{12, 8, 15, 6, 10, 14, 9, 11\}$$

Part (a)

Calculate the Fisher Information $I(\lambda)$ for a single observation from an Exponential(λ) distribution. Show your derivation using the definition $I(\lambda) = E[-\ell''(\lambda)]$.

For $X \sim \text{Exponential}(\lambda)$, the pdf is $f(x; \lambda) = \lambda e^{-\lambda x}$ for $x > 0$.

Log-likelihood for a single observation:

$$\ell(\lambda) = \ln f(x; \lambda) = \ln \lambda - \lambda x$$

First derivative:

$$\ell'(\lambda) = \frac{1}{\lambda} - x$$

Second derivative:

$$\ell''(\lambda) = -\frac{1}{\lambda^2}$$

Fisher Information:

$$I(\lambda) = E[-\ell''(\lambda)] = E\left[\frac{1}{\lambda^2}\right] = \boxed{\frac{1}{\lambda^2}}$$

Note: Since the second derivative does not depend on X , the expectation is straightforward.

Part (b)

State the Cramér-Rao Lower Bound (CRLB) for unbiased estimators of λ based on a sample of size n .

For n observations, the Fisher Information is:

$$I_n(\lambda) = n \cdot I(\lambda) = \frac{n}{\lambda^2}$$

The **Cramér-Rao Lower Bound** for unbiased estimators of λ is:

$$\boxed{\text{CRLB} = \frac{1}{I_n(\lambda)} = \frac{\lambda^2}{n}}$$

Part (c)

The MLE of λ is $\hat{\lambda} = 1/\bar{X}$. This estimator is biased. Find an unbiased estimator of λ and compute its variance. Does this unbiased estimator achieve the CRLB?

[JM NOTE: Apologies! I meant to give you the “hint” about the distribution of $1/Y$ used below and only realized I hadn’t put it in the question when one of you asked me a question about it. Note that $\bar{X}$ is unbiased for $1/\lambda$ but $1/\bar{X}$ is not unbiased for λ since $g(x) = 1/x$ is a convex function.]

For Exponential(λ), we have $\sum_{i=1}^n X_i \sim \text{Gamma}(n, \lambda)$.

Using properties of the Gamma distribution, for $Y \sim \text{Gamma}(n, \lambda)$:

$$E\left(\frac{1}{Y}\right) = \frac{\lambda}{n-1} \quad \text{for } n > 1$$

Therefore:

$$E\left(\frac{1}{\bar{X}}\right) = E\left(\frac{n}{\sum X_i}\right) = \frac{n\lambda}{n-1}$$

So the MLE is biased with $E(\hat{\lambda}) = \frac{n}{n-1}\lambda > \lambda$.

Unbiased estimator:

$$\boxed{\hat{\lambda}_U = \frac{n-1}{n} \cdot \frac{1}{\bar{X}} = \frac{n-1}{\sum_{i=1}^n X_i}}$$

Variance of the unbiased estimator:

For $Y \sim \text{Gamma}(n, \lambda)$, we have:

$$\text{Var}\left(\frac{1}{Y}\right) = \frac{\lambda^2}{(n-1)^2(n-2)} \quad \text{for } n > 2$$

Therefore:

$$\text{Var}(\hat{\lambda}_U) = (n-1)^2 \cdot \text{Var}\left(\frac{1}{\sum X_i}\right) = (n-1)^2 \cdot \frac{\lambda^2}{(n-1)^2(n-2)} = \boxed{\frac{\lambda^2}{n-2}}$$

Comparison with CRLB:

$$\text{Var}(\hat{\lambda}_U) = \frac{\lambda^2}{n-2} > \frac{\lambda^2}{n} = \text{CRLB}$$

The unbiased estimator does **NOT** achieve the CRLB.

Part (d)

Using **R**, compute the MLE $\hat{\lambda}$ from the pilot study data. Also compute the CRLB at this estimated value. Interpret what this bound tells us about estimation precision.

```
# Pilot study data
bp_times <- c(12, 8, 15, 6, 10, 14, 9, 11)
n <- length(bp_times)

# MLE
lambda_mle <- 1 / mean(bp_times)

# Unbiased estimator
lambda_unbiased <- (n - 1) / sum(bp_times)

# CRLB at the MLE estimate
crlb <- lambda_mle^2 / n

cat("Sample size: n =", n, "\n")
```

Sample size: n = 8

```
cat("Sample mean:", mean(bp_times), "days\n")
```

Sample mean: 10.625 days

```
cat("MLE of :", round(lambda_mle, 4), "per day\n")
```

MLE of : 0.0941 per day

```
cat("Unbiased estimate of :", round(lambda_unbiased, 4), "per day\n")
```

Unbiased estimate of : 0.0824 per day

```
cat("CRLB at MLE:", round(crlb, 6), "\n")
```

CRLB at MLE: 0.001107

```
cat("Standard error bound:", round(sqrt(crlb), 4), "\n")
```

Standard error bound: 0.0333

Interpretation: The CRLB tells us that no unbiased estimator of λ can have variance smaller than 0.001107. This means the standard error of any unbiased estimator must be at least 0.0333. Our estimate of $\lambda \approx 0.0941$ per day (corresponding to a mean time of 10.6 days) has inherent uncertainty—we can expect our estimates to vary by roughly ± 0.067 across repeated samples of this size.

Problem 2: Method of Moments (Shifted Exponential Distribution)

The time (in hours) from hospital admission to discharge for patients undergoing a minor outpatient procedure can be modeled with a shifted (two-parameter) Exponential distribution. This accounts for a minimum required observation time θ plus additional random waiting time. The pdf is:

$$f(x; \lambda, \theta) = \lambda e^{-\lambda(x-\theta)}, \quad x \geq \theta, \quad \lambda > 0, \quad \theta > 0$$

For this distribution:

$$E(X) = \theta + \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2}$$

Part (a)

Derive the Method of Moments estimators $\tilde{\lambda}$ and $\tilde{\theta}$ in terms of the sample mean $\bar{X}$ and sample standard deviation S .

For the shifted Exponential distribution, the population moments are:

$$E(X) = \theta + \frac{1}{\lambda}, \quad E(X^2) = \text{Var}(X) + [E(X)]^2 = \frac{1}{\lambda^2} + \left(\theta + \frac{1}{\lambda}\right)^2$$

The corresponding sample moments are:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i, \quad \frac{1}{n} \sum_{i=1}^n X_i^2$$

Method of Moments: Set population moments equal to sample moments.

First moment equation:

$$\begin{aligned} E(X) &= \bar{X} \\ \theta + \frac{1}{\lambda} &= \bar{X} \end{aligned} \quad (1)$$

Second moment equation:

$$E(X^2) = \frac{1}{n} \sum_{i=1}^n X_i^2$$

It is easier to work with the variance. Recall that $\text{Var}(X) = E(X^2) - [E(X)]^2$, so:

$$E(X^2) - [E(X)]^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2$$

The left side is $\text{Var}(X) = \frac{1}{\lambda^2}$, and the right side is the sample variance (with n in denominator):

$$\frac{1}{\lambda^2} = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \quad (2)$$

Note that the sample standard deviation is $S = \sqrt{\frac{1}{n-1} \sum (X_i - \bar{X})^2}$, so:

$$\frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{n-1}{n} S^2$$

Substituting into equation (2):

$$\frac{1}{\lambda^2} = \frac{n-1}{n} S^2 \quad (3)$$

Solving the system:

Step 1: From equation (3), solve for λ :

$$\frac{1}{\lambda^2} = \frac{n-1}{n} S^2 \implies \lambda = \sqrt{\frac{n}{n-1}} \cdot \frac{1}{S}$$

Step 2: Substitute into equation (1) to find θ :

$$\theta + \frac{1}{\lambda} = \bar{X}$$

$$\theta + S\sqrt{\frac{n-1}{n}} = \bar{X}$$

$$\theta = \bar{X} - S\sqrt{\frac{n-1}{n}}$$

Method of Moments Estimators:

$$\tilde{\lambda} = \sqrt{\frac{n}{n-1}} \cdot \frac{1}{S} = \frac{1}{S} \sqrt{\frac{n}{n-1}}$$

$$\tilde{\theta} = \bar{X} - S\sqrt{\frac{n-1}{n}}$$

Part (b)

Using R, calculate the MME estimates $\tilde{\lambda}$ and $\tilde{\theta}$. Verify that your estimate of θ is less than the minimum observed value.

```
# Discharge times (hours)
discharge_times <- c(3.2, 4.1, 2.8, 5.3, 3.5, 4.8, 3.1, 6.2, 4.4, 3.9)
n <- length(discharge_times)

# Sample statistics
xbar <- mean(discharge_times)
s <- sd(discharge_times) # This is S with (n-1) denominator

# Method of Moments estimates (exact formulas)
```

```
lambda_mme <- (1/s) * sqrt(n / (n - 1))
theta_mme <- xbar - s * sqrt((n - 1) / n)

cat("Sample size: n =", n, "\n")
```

Sample size: n = 10

```
cat("Sample mean:", round(xbar, 4), "hours\n")
```

Sample mean: 4.13 hours

```
cat("Sample SD (S):", round(s, 4), "hours\n\n")
```

Sample SD (S): 1.0709 hours

```
cat("MME estimates (exact formulas):\n")
```

MME estimates (exact formulas):

```
cat(" ~ =", round(lambda_mme, 4), "per hour\n")
```

~ = 0.9843 per hour

```
cat(" ~ =", round(theta_mme, 4), "hours\n\n")
```

~ = 3.1141 hours

```
# Validity check
min_obs <- min(discharge_times)
cat("Validity check:\n")
```

Validity check:

```
cat(" Minimum observed value:", min_obs, "hours\n")
```

Minimum observed value: 2.8 hours

```
cat(" ~ =", round(theta_mme, 4), "hours\n")
```

~ = 3.1141 hours

```
cat(" Is ~ < min(X)?", theta_mme < min_obs, "\n\n")
```

Is ~ < min(X)? FALSE

```
# Interpretation  
cat("Interpretation:\n")
```

Interpretation:

```
cat(" Estimated minimum observation time:", round(theta_mme, 2), "hours\n")
```

Estimated minimum observation time: 3.11 hours

```
cat(" Estimated mean additional wait time (1/):", round(1/lambda_mme, 2), "hours\n")
```

Estimated mean additional wait time (1/): 1.02 hours

```
cat(" Estimated total mean discharge time:", round(theta_mme + 1/lambda_mme, 2), "hours\n")
```

Estimated total mean discharge time: 4.13 hours

```
cat(" (Check: this should equal  $\bar{x}$  =", round(xbar, 2), "hours)\n")
```

(Check: this should equal $\bar{x}$ = 4.13 hours)

Interpretation: The MME estimates suggest that patients require a minimum of approximately 3.11 hours of observation, with an additional expected random waiting time of 1.02 hours before discharge. The validity check confirms that $\tilde{\theta}$ is less than the smallest observed discharge time (2.8 hours), which is necessary for the model to be consistent with the data.

Problem 3: Maximum Likelihood Estimation (Devore 7.25)

Let $X_1, X_2, \dots, X_n$ be a random sample from a distribution with pdf:

$$f(x; \theta) = (\theta + 1)x^\theta, \quad 0 \leq x \leq 1, \quad \theta > -1$$

Part (a)

Derive the maximum likelihood estimator $\hat{\theta}$ of θ .

Likelihood:

$$L(\theta) = \prod_{i=1}^n (\theta + 1)x_i^\theta = (\theta + 1)^n \prod_{i=1}^n x_i^\theta = (\theta + 1)^n \left(\prod_{i=1}^n x_i \right)^\theta$$

Log-likelihood:

$$\ell(\theta) = n \ln(\theta + 1) + \theta \sum_{i=1}^n \ln(x_i)$$

First derivative:

$$\ell'(\theta) = \frac{n}{\theta + 1} + \sum_{i=1}^n \ln(x_i)$$

Setting $\ell'(\theta) = 0$:

$$\begin{aligned} \frac{n}{\theta + 1} &= - \sum_{i=1}^n \ln(x_i) \\ \theta + 1 &= \frac{n}{-\sum_{i=1}^n \ln(x_i)} = \frac{-n}{\sum_{i=1}^n \ln(x_i)} \\ \theta &= \frac{-n}{\sum_{i=1}^n \ln(x_i)} - 1 \end{aligned}$$

MLE:

$$\hat{\theta} = \frac{-n}{\sum_{i=1}^n \ln(x_i)} - 1 = -1 - \frac{1}{\overline{\ln(X)}}$$

where $\overline{\ln(X)} = \frac{1}{n} \sum_{i=1}^n \ln(x_i)$.

Verification (second derivative test):

$$\ell''(\theta) = -\frac{n}{(\theta + 1)^2} < 0$$

This confirms we have a maximum.

Part (b)

For a sample of size $n = 5$ with observations $x_1 = 0.92, x_2 = 0.79, x_3 = 0.85, x_4 = 0.95, x_5 = 0.88$, compute the MLE $\hat{\theta}$.

```
x <- c(0.92, 0.79, 0.85, 0.95, 0.88)
n <- length(x)

# MLE calculation
sum_ln_x <- sum(log(x))
theta_mle <- -n / sum_ln_x - 1

cat("Observations:", x, "\n")
```

Observations: 0.92 0.79 0.85 0.95 0.88

```
cat("Sum of ln(x_i):", round(sum_ln_x, 4), "\n")
```

Sum of ln(x_i): -0.6607

```
cat("MLE ^ =", round(theta_mle, 4), "\n")
```

MLE ^ = 6.5672

Part (c)

Use the invariance property of MLEs to find the MLE of $\tau(\theta) = \frac{\theta+1}{\theta+2}$, which represents $E(X)$ for this distribution.

By the invariance property of MLEs, the MLE of $\tau(\theta) = g(\theta)$ is $\hat{\tau} = g(\hat{\theta})$.

Therefore:

$$\hat{\tau} = \frac{\hat{\theta} + 1}{\hat{\theta} + 2}$$

```
tau_mle <- (theta_mle + 1) / (theta_mle + 2)
cat("MLE of ( ) = (+1)/(+2):", round(tau_mle, 4), "\n")
```

MLE of () = (+1)/(+2): 0.8833

```
cat("Sample mean (for comparison):", round(mean(x), 4), "\n")
```

Sample mean (for comparison): 0.878

Note: The MLE of $E(X)$ (0.8833) is close to but not exactly equal to the sample mean (0.878). This is expected since the MLE uses the full likelihood structure of the distribution.

Problem 4: Sufficient Statistics (Medical Application)

A researcher is studying the number of adverse events reported per patient in a drug trial. The number of adverse events X follows a $\text{Poisson}(\mu)$ distribution.

Part (a)

For a random sample $X_1, \dots, X_n$ from $\text{Poisson}(\mu)$, use the Neyman Factorization Theorem to show that $T = \sum_{i=1}^n X_i$ is a sufficient statistic for μ .

For $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Poisson}(\mu)$, the joint pmf is:

$$f(x_1, \dots, x_n; \mu) = \prod_{i=1}^n \frac{e^{-\mu} \mu^{x_i}}{x_i!} = \frac{e^{-n\mu} \mu^{\sum_{i=1}^n x_i}}{\prod_{i=1}^n x_i!}$$

We can factor this as:

$$f(x_1, \dots, x_n; \mu) = \underbrace{e^{-n\mu} \mu^{\sum x_i}}_{g(T; \mu)} \cdot \underbrace{\frac{1}{\prod_{i=1}^n x_i!}}_{h(x_1, \dots, x_n)}$$

where $T = \sum_{i=1}^n X_i$.

- $g(T; \mu) = e^{-n\mu} \mu^T$ depends on the data **only through** T and involves μ
- $h(x_1, \dots, x_n) = 1 / \prod x_i!$ depends only on the data, **not on** μ

By the **Neyman Factorization Theorem**, $T = \sum_{i=1}^n X_i$ is **sufficient for** μ . $\square$

Part (b)

Show that the MLE $\hat{\mu} = \bar{X}$ is a function of the sufficient statistic T .

The MLE is:

$$\hat{\mu} = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i = \frac{T}{n}$$

Since $\hat{\mu} = T/n$, the MLE can be written purely as a function of the sufficient statistic T . ✓

Part (c)

In a clinical trial with $n = 50$ patients, suppose the following summary statistics are recorded:

$$\sum_{i=1}^{50} X_i = 72$$

A colleague suggests using a different estimator: $\tilde{\mu} = \frac{\text{number of patients with } X_i > 0}{n}$.

Using R simulation, generate 1000 samples from $\text{Poisson}(\mu = 1.5)$ with $n = 50$ and compare the bias, variance and MSE of $\hat{\mu}$ and $\tilde{\mu}$.

[JM side note: You don't need to use `map()` functions like I do, unless you want to learn them! Feel free to make this work however you know you to make tibbles (stack, make each vector separately, whatever).]

```
set.seed(551)
n <- 50
mu_true <- 1.5
n_sims <- 1000

# Simulation
sim_results <- tibble(sim = 1:n_sims) |>
  mutate(
    data = map(sim, \(s) rpois(n, mu_true)),
    mu_hat = map_dbl(data, mean), # MLE (based on sufficient stat)
    mu_tilde = map_dbl(data, \(x) mean(x > 0)) # Proportion with events
  )

# Compute summaries
summary_stats <- sim_results |>
  summarize(
    Estimator = c("̂ = X̄ (MLE)", "̃ = proportion(X > 0)"),
```

```

    True_Value = c(mu_true, 1 - exp(-mu_true)), # Note:  $E(\tilde{\mu}) = P(X > 0) = 1 - e^{-\mu}$ 
    Mean = c(mean(mu_hat), mean(mu_tilde)),
    Bias = c(mean(mu_hat) - mu_true, mean(mu_tilde) - mu_true),
    Variance = c(var(mu_hat), var(mu_tilde)),
    MSE = c(mean((mu_hat - mu_true)^2), mean((mu_tilde - mu_true)^2))
  )

print(summary_stats)

```

A tibble: 2 x 6

	Estimator	True_Value	Mean	Bias	Variance	MSE
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	$\hat{\mu} = \bar{X}$ (MLE)	1.5	1.50	0.00384	0.0309	0.0309
2	$\tilde{\mu} = \text{proportion}(X > 0)$	0.777	0.778	-0.722	0.00367	0.525

Visualization

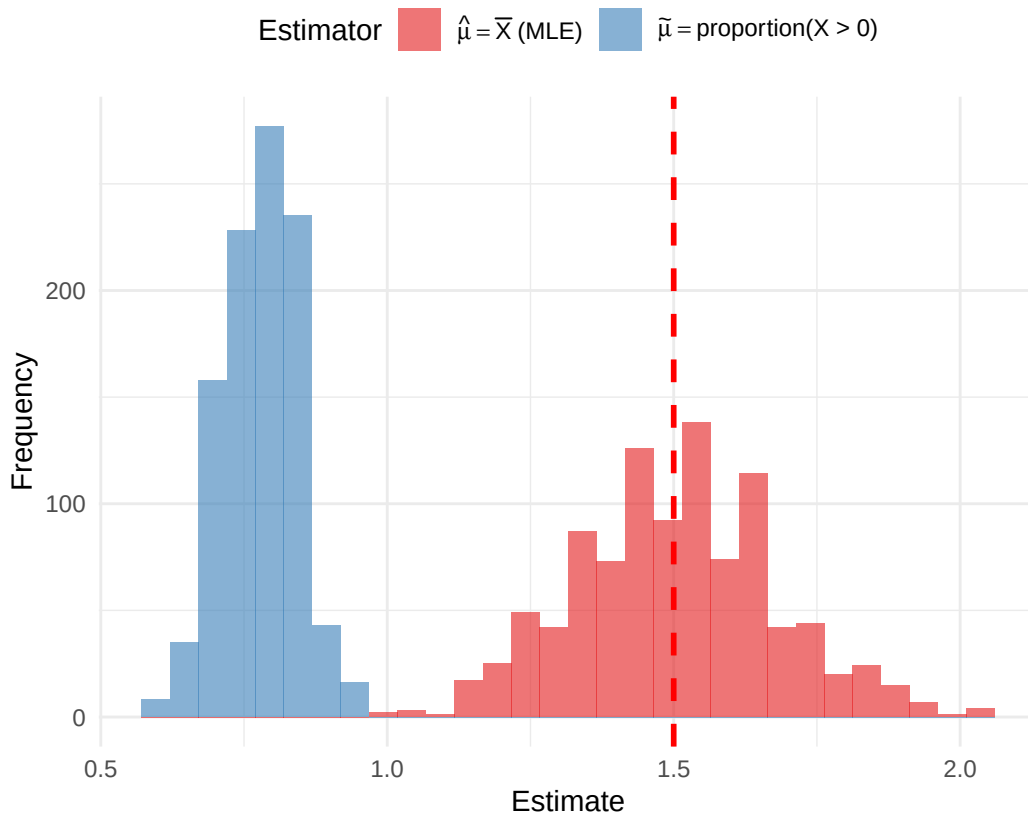
```

sim_results |>
  pivot_longer(cols = c(mu_hat, mu_tilde),
    names_to = "Estimator", values_to = "Estimate") |>
  mutate(Estimator = factor(Estimator,
    levels = c("mu_hat", "mu_tilde"),
    labels = c("MLE", "Proportion"))) |>
  ggplot(aes(x = Estimate, fill = Estimator)) +
  geom_histogram(alpha = 0.6, bins = 30, position = "identity") +
  geom_vline(xintercept = mu_true, linetype = "dashed", color = "red", linewidth = 1) +
  labs(title = bquote("Sampling Distributions of Two Estimators for" ~ mu),
    subtitle = bquote("True" ~ mu == .(mu_true) ~ "|" ~ n == .(n) ~ "| Simulations =" ~ .),
    x = "Estimate", y = "Frequency") +
  scale_fill_brewer(palette = "Set1",
    labels = c(expression(hat(mu) == bar(X) ~ "(MLE)"),
      expression(tilde(mu) == "proportion(X > 0)"))) +
  theme_minimal() +
  theme(legend.position = "top")

```

Sampling Distributions of Two Estimators for μ

True $\mu = 1.5$ | $n = 50$ | Simulations = 1000



Interpretation:

- The MLE $\hat{\mu} = \bar{X}$ is essentially unbiased with bias ≈ 0 and estimates μ directly.
- The proportion estimator $\tilde{\mu}$ is heavily biased because it estimates $P(X > 0) = 1 - e^{-\mu} \approx 0.777$, not μ itself.
- Even though $\tilde{\mu}$ has smaller variance, its large bias results in much larger MSE.
- The MLE, which is based on the sufficient statistic $T = \sum X_i$, is clearly superior for estimating μ .

Problem 5: MLE for Normal Variance (Devore 7.31 Modified)

Measurement errors in a laboratory instrument are assumed to follow a Normal distribution with known mean $\mu = 0$ and unknown variance σ^2 . A sample of n measurements yields observations $X_1, \dots, X_n$.

Part (a)

Derive the MLE of σ^2 .

For $X_i \stackrel{iid}{\sim} N(0, \sigma^2)$, the pdf is:

$$f(x; \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-x^2/(2\sigma^2)}$$

Likelihood:

$$L(\sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-x_i^2/(2\sigma^2)} = (2\pi\sigma^2)^{-n/2} \exp\left(-\frac{\sum_{i=1}^n x_i^2}{2\sigma^2}\right)$$

Log-likelihood:

$$\ell(\sigma^2) = -\frac{n}{2} \ln(2\pi) - \frac{n}{2} \ln(\sigma^2) - \frac{\sum_{i=1}^n x_i^2}{2\sigma^2}$$

Derivative with respect to σ^2 :

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{\sum_{i=1}^n x_i^2}{2(\sigma^2)^2}$$

Setting equal to zero:

$$\begin{aligned} \frac{n}{2\sigma^2} &= \frac{\sum_{i=1}^n x_i^2}{2(\sigma^2)^2} \\ n\sigma^2 &= \sum_{i=1}^n x_i^2 \end{aligned}$$

MLE:

$$\boxed{\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n X_i^2}$$

Part (b)

Is this MLE unbiased? If not, find the bias.

Since $X_i \sim N(0, \sigma^2)$, we have:

$$E(X_i^2) = \text{Var}(X_i) + [E(X_i)]^2 = \sigma^2 + 0 = \sigma^2$$

Therefore:

$$E(\hat{\sigma}^2) = E\left(\frac{1}{n} \sum_{i=1}^n X_i^2\right) = \frac{1}{n} \cdot n \cdot \sigma^2 = \sigma^2$$

The MLE is **unbiased**:

$$\text{Bias}(\hat{\sigma}^2) = E(\hat{\sigma}^2) - \sigma^2 = 0$$

Part (c)

A technician takes $n = 15$ measurements and records: $\sum_{i=1}^{15} X_i^2 = 4.8$

Calculate the MLE of σ^2 and provide an unbiased estimate.

```
n <- 15
sum_x_squared <- 4.8

# MLE (which is also unbiased in this case)
sigma2_mle <- sum_x_squared / n

cat("n =", n, "\n")
```

n = 15

```
cat("Sum of  $X_i^2$  =", sum_x_squared, "\n\n")
```

Sum of X_i^2 = 4.8

```
cat("MLE of  $\sigma^2$ :", round(sigma2_mle, 4), "\n")
```

MLE of σ^2 : 0.32

```
cat("Unbiased estimate of  $\sigma^2$ :", round(sigma2_mle, 4), "(same as MLE)\n")
```

Unbiased estimate of σ^2 : 0.32 (same as MLE)

```
cat("Estimated  $\sigma$ :", round(sqrt(sigma2_mle), 4), "\n")
```

Estimated σ : 0.5657

Note: In this special case where $\mu = 0$ is known, the MLE $\hat{\sigma}^2 = \frac{1}{n} \sum X_i^2$ is already unbiased. This is different from the usual case where μ is unknown, in which case the MLE $\frac{1}{n} \sum (X_i - \bar{X})^2$ is biased and we use $\frac{1}{n-1} \sum (X_i - \bar{X})^2$ for an unbiased estimate.